

Arogya

Ayurvedic Based; Disease Prediction, Medicine Recommendation, Diet Planning with Smart Q&A System

- People in rural areas often cannot access qualified Ayurvedic doctors due to long distances and high costs.
- Many rely on self-medication or generic online advice, which can be unsafe.
- 91.1% of Sri Lankans prefer Ayurvedic remedies at their fingertips (Survey, n=235).

Affected Parties:

- **Patients:** Delayed diagnosis, inconsistent treatments, lack of personalized care
- **Rural Communities:** Limited access to specialized Ayurvedic knowledge
- **Traditional Practitioners:** Need for decision support tools

Importance of Solving This Problem

Immediate Healthcare Impact

- Prevent unsafe self-medication affecting millions
- Reduce travelling cost and

• Bridge urban-rural healthcare divide



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Ayurvedic Based Disease Prediction

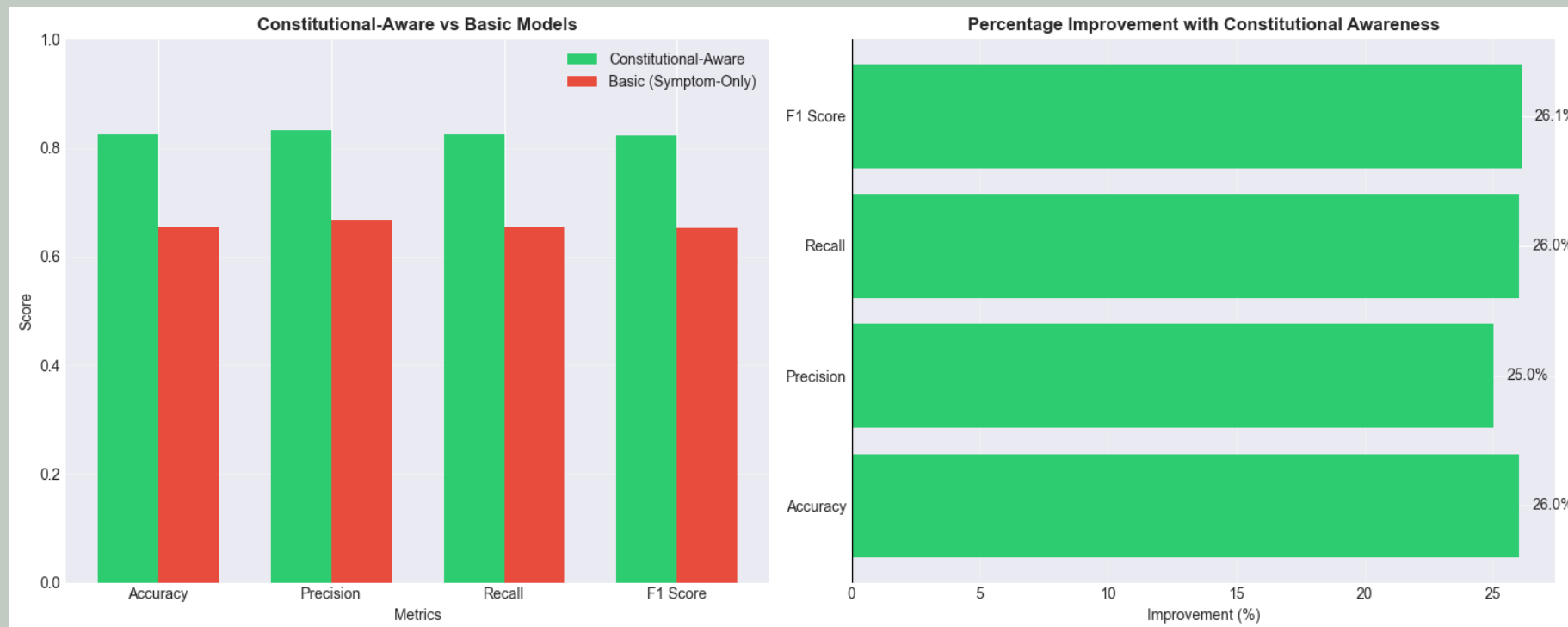
Specialization: Information Technology

AYURVEDIC BASED DISEASE PREDICTION

- Same symptoms indicate different diseases based on individual constitution (Prakriti)
- 6-question Prakriti assessment, symptom severity tracking, instant AI predictions, disease information with recommendations
- Created first-of-its-kind dataset (1000 patients, 2992 records) with complete dosha coverage from classical texts (Charaka Samhita, Madhava Nidana)
- React+Flask web interface with real-time constitutional assessment and ML prediction.

AYURVEDIC BASED DISEASE PREDICTION

- Novel Contribution: First ML system to integrate Prakriti (constitutional type) with symptom analysis - demonstrating 10% accuracy improvement
- Technical Achievement: Random Forest classifier achieving 84.2% accuracy across 5 Ayurvedic diseases with balanced performance





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Medicine recommendation module

Specialization: Information Technology

AYUVEDIC HERB RECOMMENDATION

•Problem Statement:

- Manual Ayurvedic herb selection is time-consuming (2-3 minutes per patient), inconsistent across practitioners, and lacks empirical validation of traditional dosha-based diagnostic principles.

Impact: Transforms subjective, time-intensive herb selection into data-driven, efficient **clinical decision support** while preserving traditional Ayurvedic wisdom and maintaining expert clinical judgment as the final authority.

•Design Excellence:

- **Simplicity:** Only 4 input fields
- **Visual Hierarchy:** Best recommendation highlighted separately
- **Progressive Disclosure:** Alternatives shown after best recommendation
- **Feedback:** Loading animation + instant results
- **Safety:** Clear disclaimer about practitioner consultation

AYURVEDIC HERB RECOMMENDATION

- **Novel Contributions:**

- **First systematic ML study** for 15-class Ayurvedic herb recommendation
- **Top-K evaluation framework** aligned with clinical workflows
- **Empirical dosha validation:** Quantified that dosha adds predictive value
- **Algorithm-independent validation:** Three models converge → robust patterns

- **Research Methodology:**

- **Data Collection:** 2,000 patient-herb cases with 4 clinical features
- **Preprocessing:** OneHotEncoding, Pipeline architecture (prevent data leakage)
- **Modeling:** Tested 3 algorithms (RF, DT, SVM)
- **Evaluation:** Top-K accuracy, 5-fold cross-validation, per-herb analysis



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AYURVEDIC BASED MEAL PLAN RECOMMENDATION

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AYURVEDIC BASED MEAL PLAN RECOMMENDATION

•Identified Sub-Problem

- Lack of personalized, Ayurveda-based dietary guidance; they are not dosha based, with lack of nutrient values.**

Proposed Solution (Prototype Overview)

The proposed system is a **Personalized Ayurvedic Meal Recommendation System** that uses a **machine learning-based approach** to generate meal plans.

The prototype:

- Accepts user-specific inputs
- Predicts the dominant Dosha
- Recommends a **complete, balanced meal**
- Provides nutritional summaries
- Suggests foods to avoid based on the condition

User Requirements addressed by the solution

- Personalized dietary guidance aligned with **Ayurvedic principles**
- Helps users make **informed food choices**
- Supports better disease management through nutrition
- Acts as a **decision-support tool**, not a replacement for medical professionals

AYURVEDIC BASED MEAL PLAN RECOMMENDATION

The proposed solution demonstrates design excellence through:

- Integration of **traditional Ayurvedic dietary principles** with **modern machine learning techniques**
- A structured, modular system design separating:
 - Data preparation
 - Model training and evaluation
 - User interface
- Use of a **multi-label recommendation approach** to generate complete meals rather than isolated food items
- **Feedback:**
 - User interface is clean, simple, and easy to navigate
 - Meal recommendations are easy to understand and feel practical
 - Nutrient summary adds clarity and trust to the recommendations
 - Dosha information helps users understand their body type



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AYURVEDIC BASED QUESTION ANSWERING SYSTEM

Specialization: INFORMATION TECHNOLOGY

AYURVEDIC BASED QUESTION ANSWERING SYSTEM

•Identified Sub-Problem

•Converts inaccessible Ayurvedic knowledge (scanned books, unstructured texts) into an intelligent, searchable Q&A system using OCR → Dataset Creation → RAG Architecture

Proposed Solution (Prototype Overview)

The proposed system is an **AI-Powered Ayurvedic Question-Answering System** that converts unstructured classical texts into structured, searchable knowledge using NLP and vector embeddings.

- **Processes scanned Ayurvedic books** via OCR and text preprocessing
- **Auto-generates Q&A dataset** from extracted text using LLM-based analysis
- **Stores semantic embeddings** in vector database for intelligent retrieval
- **Implements RAG architecture** combining retrieval + generation for authentic answers
- **Ensures traceability** with complete citation metadata for every response

User Requirements addressed by the solution

- Instant access to authentic Ayurvedic knowledge** with verified citations from classical texts
- Bridges knowledge gap** between ancient terminology and modern understanding through dual question formats
- Enables evidence-based learning** - students and practitioners can trace answers back to original sources
- Acts as an educational and reference tool**, not a replacement for Ayurvedic physicians or clinical judgment

AYURVEDIC BASED MEAL PLAN RECOMMENDATION

The proposed solution demonstrates design excellence through:

- **Integration of ancient Ayurvedic knowledge with cutting-edge AI technologies** (RAG, LLMs, Vector Databases)
- **A structured, end-to-end pipeline design** separating:
 - OCR and text extraction
 - Automated dataset creation
 - Embedding generation and vector storage
 - RAG-based retrieval and answer generation
- **Novel automated Q&A generation approach** creating structured datasets from unstructured classical texts
- **Citation preservation system** ensuring every answer is traceable to original source (book, chapter, paragraph)
- **Semantic search implementation** using vector embeddings for meaning-based retrieval, not just keyword matching
- **User Feedback on Prototype**
- **Interface is intuitive and straightforward** - users can ask questions naturally without technical knowledge
- **Answers are clear, concise, and contextually relevant** to the query
- **Source citations build trust and credibility** - users appreciate ability to verify information
- **Dual question format support** helps both beginners and Ayurvedic experts find relevant information
- **Fast response times** make it practical for real-time consultation support

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Commercialization/Sustainability of Solution

- Customer Persona: Health-conscious individuals (25-55 years), rural communities with limited practitioner access, Ayurvedic clinics seeking decision support tools
- Sri Lankan Ayurvedic market valued at 450M+, with 70% population preferring traditional medicine for common ailments